Modeling Predictors of Smoking Pack–Years Among U.S. Adults: A Multiple Regression Analysis Using NHIS Data

Alexandra Chang

November 24, 2025

Abstract

This project uses data from the 2020 National Health Interview Survey to model cigarette pack–years using demographic, behavioral, and comorbidity predictors. After excluding cases with missing smoking history variables, 11,300 adults remained for complete-case analysis. A multiple linear regression model was fit using smoking duration, cigarettes per day, and comorbidities as predictors. Model selection using backward, forward, stepwise AIC/BIC criteria and all-subset regression consistently selected a reduced model including age, race, smoking duration, cigarettes per day, BMI, and several comorbidities. The final model explained 85.2% of the variation in pack–years. Diagnostic plots revealed substantial heteroscedasticity, heavy-tailed residuals, and nonlinearity for smoking-related predictors, though influence statistics indicated no harmful high-leverage or influential observations. These results highlight the dominant contribution of smoking intensity and duration to lifetime tobacco exposure and demonstrate limitations of linear modeling for highly skewed behavioral outcomes.

1 Introduction

This project applies multiple linear regression methods covered in STAT 525 (Intermediate Statistical Methodology) to a large health survey dataset derived from the National Health Interview Survey (NHIS). The primary research goal is to model *pack–years of smoking* (pky) as a function of demographic factors, smoking behaviors, and comorbidities. Pack–years is a standard epidemiologic measure that combines cigarettes per day and duration of smoking, and serves as an important risk indicator in clinical and public health applications.

The dataset was constructed by recoding relevant NHIS variables, removing respondents younger than 18, and excluding observations missing essential smoking information. After cleaning, the final analytic sample included approximately 11,300 respondents with complete data for regression analysis.

Using methods taught in class—including fitting a full model, conducting backward, forward, and stepwise selection, evaluating AIC/BIC, and performing regression diagnostics—we assess which variables significantly contribute to predicting pack–years and whether the assumptions of the linear model are satisfied.

The remainder of this report is structured as follows. Section 2 describes the data preparation procedures and statistical methods. Section 3 presents regression results and model selection. Section 4 provides diagnostic analysis to evaluate the validity of the linear model. Section 5 concludes with a summary of findings and limitations. All R code used in this project is included in the Appendix.

2 Methods

2.1 Data Source and Study Sample

The data used in this project come from the publicly available 2020 National Health Interview Survey (NHIS), a nationally representative survey of U.S. adults. From the full NHIS dataset, only respondents aged 18 years or older were considered. Individuals missing key smoking variables (such as smoking duration, cigarettes per day, or derived pack–years) were excluded. After applying these restrictions and recoding steps, the final analytic dataset consisted of 11,300 complete observations. We used complete-case analysis because imputation was not required for STAT 525 and because key smoking variables were missing for most excluded observations.

2.2 Variable Recoding and Derived Measures

All recoding was performed in R. Several NHIS variables required harmonization or derivation before analysis:

- **Smoking duration (smkyears):** For current smokers, duration was calculated as age minus age at smoking initiation. For former smokers, duration was calculated as age minus initiation age minus years since quitting. For never smokers, the value was set to zero.
- **Average cigarettes per day (avecpd):** Combined information from multiple NHIS fields (CIGSDAY1, CIGSDAY2, CIGSDAYFS) to obtain a single measure of typical daily cigarette use. Never smokers were assigned zero.
- **Pack–years (pky):** Defined as:

$$\text{pky} = \frac{\text{cigarettes per day} \times \text{smoking duration (years)}}{20}.$$

  Negative or nonsensical values were set to missing.
- **Demographics:** Age, sex, race/ethnicity, and education level were standardized into numeric or factor formats appropriate for regression.
- **Comorbidities:** Chronic disease indicators (e.g., COPD, diabetes, hypertension, heart disease) were recoded as binary variables reflecting the presence or absence of the condition.
- **Family history of cancer (fmhist):** Derived from NHIS variables on parental cancer history.

These recoding steps ensured all predictors were usable in linear regression and consistent across the full sample.

2.3 Statistical Model

We modeled pack–years of smoking as a continuous outcome using multiple linear regression. The initial *full model* included all demographic, behavioral, and comorbidity predictors:

$$\text{pky} = \beta_0+\beta_1\text{age}+\beta_2\text{female}+\beta_3\text{race}+\beta_4\text{edu}+\beta_5\text{smkyears}+\beta_6\text{avecpd}+\beta_7\text{qtyears}+\beta_8\text{bmi}+\beta_9\text{copd}+\cdots+\beta_{17}\text{fmhist}+\varepsilon.$$

Categorical predictors (race, education) were included using R's default treatment contrasts.

2.4 Model Selection Procedures

Because many predictors were available, several model selection techniques from STAT 525 were applied:

- **Backward elimination** using AIC.
- **Forward selection** starting from the intercept-only model.
- **Stepwise selection** allowing both addition and removal of predictors.
- **Comparison by AIC and BIC** across all three procedures and the full model.
- **All–subset regression (regsubsets)** to examine adjusted R^2 and Mallows' C_p across models of different sizes (after converting categorical variables to numeric codes for compatibility).

All model selection was performed using functions `stepAIC()` from the `MASS` package and `regsubsets()` from the `leaps` package.

Across procedures, the backward, forward, and stepwise models all converged to the same reduced model, which was chosen as the *final regression model*.

2.5 Regression Diagnostics

To assess whether the assumptions of the linear model were satisfied, we examined:

- Residuals vs. fitted values (linearity & homoscedasticity)
- Residuals vs. each continuous and binary predictor
- Normal Q–Q plot of residuals
- Histogram of residuals
- Leverage values (hat matrix diagonal)
- Cook's distance
- DFFITS
- Studentized deleted residuals

These diagnostics were generated using base R functions and `ggplot2`. Values exceeding commonly used thresholds (e.g., Cook's $D > 4/n$, DFFITS $> 2\sqrt{p/n}$, or studentized residuals exceeding ± 3) were noted but not removed, as the goal was to evaluate model behavior rather than refine the dataset.

3 Results

3.1 Descriptive statistics

Although the original NHIS extract contained 333,374 respondents, only 11,300 adults had complete data on all variables used in the regression model. All analyses below were conducted on this complete-case analytic dataset. After recoding and restricting the dataset to adults with complete smoking information, the final analytic sample contained $n = 11{,}300$ observations. The response variable, pack–years (pky), displayed a highly right-skewed distribution: 50% of respondents had 0 pack–years (never smokers), while the upper tail extended to nearly 300 pack–years. Smoking duration (smkyears) ranged from 0 to 78 years, and average cigarettes per day ranged from 0 to 95. Binary comorbidity indicators (e.g., COPD, diabetes, hypertension, kidney disease) had low prevalence, consistent with the general population nature of the NHIS sample.

```
      pky              age            female            race             edu           smkyears          avecpd          qtyears
 Min.   :  0.000   Min.   :18.00   Min.   :0.0000   Min.   :0.000   Min.   :1.000   Min.   : 0.000   Min.   : 0.000   Min.   : 0.000
 1st Qu.:  0.000   1st Qu.:33.00   1st Qu.:0.0000   1st Qu.:0.000   1st Qu.:2.000   1st Qu.: 0.000   1st Qu.: 0.000   1st Qu.: 0.000
 Median :  0.000   Median :48.00   Median :1.0000   Median :0.000   Median :4.000   Median : 0.000   Median : 0.000   Median : 0.000
 Mean   :  4.099   Mean   :48.61   Mean   :0.5705   Mean   :0.701   Mean   :3.488   Mean   : 5.828   Mean   : 2.849   Mean   : 5.135
 3rd Qu.:  0.000   3rd Qu.:63.00   3rd Qu.:1.0000   3rd Qu.:1.000   3rd Qu.:5.000   3rd Qu.: 0.000   3rd Qu.: 0.000   3rd Qu.: 0.000
 Max.   :289.750   Max.   :85.00   Max.   :1.0000   Max.   :3.000   Max.   :6.000   Max.   :78.000   Max.   :95.000   Max.   :70.000
                                   NA's   :9        NA's   :6491    NA's   :1427                                      NA's   :260132
      bmi             copd            diab            heartattack        heartdisease       hypertension         stroke             kidney
 Min.   :14.30   Min.   :0.000   Min.   :0.00000   Min.   :0.00000   Min.   :0.000     Min.   :0.0000   Min.   :0.00000   Min.   :0.000
 1st Qu.:23.50   1st Qu.:0.000   1st Qu.:0.00000   1st Qu.:0.00000   1st Qu.:0.000     1st Qu.:0.0000   1st Qu.:0.00000   1st Qu.:0.000
 Median :26.60   Median :0.000   Median :0.00000   Median :0.00000   Median :0.000     Median :0.0000   Median :0.00000   Median :0.000
 Mean   :27.53   Mean   :0.037   Mean   :0.09521   Mean   :0.02911   Mean   :0.073     Mean   :0.3185   Mean   :0.02932   Mean   :0.022
 3rd Qu.:30.70   3rd Qu.:0.000   3rd Qu.:0.00000   3rd Qu.:0.00000   3rd Qu.:0.000     3rd Qu.:1.0000   3rd Qu.:0.00000   3rd Qu.:0.000
 Max.   :55.30   Max.   :1.000   Max.   :1.00000   Max.   :1.00000   Max.   :1.000     Max.   :1.0000   Max.   :1.00000   Max.   :1.000
 NA's   :30282   NA's   :50999   NA's   :234       NA's   :273       NA's   :105156    NA's   :429      NA's   :274       NA's   :70505
     liver             fmhist
 Min.   :0.000    Min.   :0.000
 1st Qu.:0.000    1st Qu.:0.000
 Median :0.000    Median :0.000
 Mean   :0.016    Mean   :0.053
 3rd Qu.:0.000    3rd Qu.:0.000
 Max.   :1.000    Max.   :2.000
 NA's   :105113   NA's   :274880
```

3.2 Full multiple regression model

The full model included demographics, smoking behaviors, BMI, comorbidities, and family history:

$$\text{pky} \sim \text{age} + \text{female} + \text{race} + \text{edu} + \text{smkyears} + \text{avecpd} + \text{qtyears} + \text{bmi} + \text{comorbidities} + \text{family history}.$$

The model explained a substantial portion of variability in pack–years (multiple $R^2 = 0.8523$, adjusted $R^2 = 0.852$), indicating that smoking duration and intensity—combined with demographic and health covariates—are strong predictors of cumulative tobacco exposure.

```
Call:
lm(formula = pky ~ age + female + race + edu + smkyears + avecpd +
    qtyears + bmi + copd + diab + heartattack + heartdisease +
    hypertension + stroke + kidney + liver + fmhist, data = stat525_project)

Residuals:
    Min      1Q  Median      3Q     Max
-57.431  -4.042  -0.420   3.747  84.262
```

```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)  -12.894693   0.604271 -21.339  < 2e-16 ***
age           -0.049654   0.016757  -2.963 0.003052 **
femaleFemale   0.099678   0.166433   0.599 0.549244
raceNH_Black  -0.541111   0.270888  -1.998 0.045790 *
raceHispanic   0.953664   0.280545   3.399 0.000678 ***
raceOther      0.934084   0.347182   2.690 0.007146 **
edu.L          0.273587   0.253966   1.077 0.281389
edu.Q          0.374369   0.240786   1.555 0.120027
edu.C         -0.271989   0.213965  -1.271 0.203688
edu^4          0.071240   0.199771   0.357 0.721392
edu^5         -0.028769   0.202446  -0.142 0.886996
smkyears       0.661350   0.016528  40.014  < 2e-16 ***
avecpd         1.356649   0.007672 176.840  < 2e-16 ***
qtyears       -0.014364   0.017205  -0.835 0.403811
bmi           -0.035773   0.015502  -2.308 0.021036 *
copd           4.257506   0.281201  15.140  < 2e-16 ***
diab           0.405084   0.264679   1.530 0.125928
heartattack    1.487394   0.353936   4.202 2.66e-05 ***
heartdisease   0.081343   0.272767   0.298 0.765546
hypertension  -0.531591   0.190225  -2.795 0.005206 **
stroke        -0.087812   0.390144  -0.225 0.821925
kidney         1.140755   0.477986   2.387 0.017022 *
liver         -0.746497   0.492690  -1.515 0.129763
fmhist         0.172137   0.298237   0.577 0.563827
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 8.629 on 11276 degrees of freedom
  (322074 observations deleted due to missingness)
Multiple R-squared:  0.8523,    Adjusted R-squared:  0.852
F-statistic:  2829 on 23 and 11276 DF,  p-value: < 2.2e-16
```

Key results from the full model:

- **Smoking duration (smkyears)** was one of the strongest predictors ($\hat{\beta} = 0.661$, $t = 40.01$, $p < 0.0001$), meaning each additional year of smoking increases pack–years by approximately 0.66.
- **Average cigarettes/day (avecpd)** had the largest effect size ($\hat{\beta} = 1.357$, $t = 176.84$, $p < 0.0001$), which is expected since pack–years mathematically depend on cigarettes per day.

- **Age** had a statistically significant but small negative coefficient ($\hat{\beta} = -0.0496$, $p = 0.003$), which reflects age-adjusted differences (not biological decline). Older never-smokers dilute the average, leading to this negative partial association.
- **Race/ethnicity** showed moderate differences:
  - NH Black: $\hat{\beta} = -0.541$, $p = 0.046$
  - Hispanic: $\hat{\beta} = 0.954$, $p = 0.0007$
  - Other: $\hat{\beta} = 0.934$, $p = 0.0071$

  Effects were modest relative to smoking variables.
- **BMI** had a small negative effect ($p = 0.021$), indicating slightly lower pack–years among higher-BMI individuals after adjusting for age and smoking characteristics.
- **COPD** showed a large and highly significant association ($\hat{\beta} = 4.26$, $t = 15.14$, $p < 0.0001$), consistent with heavy smoking histories among individuals with COPD.
- **Heart attack**, **hypertension**, and **kidney disease** were also statistically significant predictors.
- Several comorbidities (liver disease, stroke, diabetes, heart disease) and family history were not statistically significant after adjusting for the main smoking variables.

3.3 ANOVA (Type I) results

Sequential ANOVA indicated that the smoking variables were overwhelmingly the strongest contributors to model fit:

```
Analysis of Variance Table

Response: pky
                Df  Sum Sq Mean Sq    F value    Pr(>F)    
age              1  953872  953872 12810.9195 < 2.2e-16 ***
female           1   47450   47450   637.2717 < 2.2e-16 ***
race             3   98952   32984   442.9876 < 2.2e-16 ***
edu              5  167910   33582   451.0205 < 2.2e-16 ***
smkyears         1 1058678 1058678 14218.5097 < 2.2e-16 ***
avecpd           1 2496472 2496472 33528.6999 < 2.2e-16 ***
qtyears          1      72      72     0.9603  0.327134    
bmi              1     395     395     5.3105  0.021215 *  
copd             1   18982   18982   254.9375 < 2.2e-16 ***
diab             1     263     263     3.5361  0.060072 .  
heartattack      1    1338    1338    17.9714  2.26e-05 ***
heartdisease     1       3       3     0.0382  0.845084    
hypertension     1     567     567     7.6179  0.005788 ** 
stroke           1       1       1     0.0149  0.902821    
kidney           1     384     384     5.1614  0.023113 *  
liver            1     170     170     2.2899  0.130249    
fmhist           1      25      25     0.3331  0.563827    
Residuals    11276  839586      74                         
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

- avecpd: $F = 33{,}529$, $p < 0.0001$
- smkyears: $F = 14{,}219$, $p < 0.0001$

Other variables contributed much smaller increments to explained variance.

3.4 Multicollinearity

Variance Inflation Factors (GVIF) showed:

- **High multicollinearity**: smkyears (GVIF = 10.14), qtyears (GVIF = 9.83), age (GVIF = 12.74);
- **Low multicollinearity**: all other predictors (GVIF < 2).

```
                  GVIF Df GVIF^(1/(2*Df))
age          12.740771  1        3.569422
female        1.048503  1        1.023964
race          1.202269  3        1.031178
edu           1.188661  5        1.017433
smkyears     10.142427  1        3.184718
avecpd        1.217226  1        1.103280
qtyears       9.830801  1        3.135411
bmi           1.098918  1        1.048293
copd          1.226669  1        1.107551
diab          1.177986  1        1.085351
heartattack   1.129768  1        1.062905
heartdisease  1.096071  1        1.046934
hypertension  1.333686  1        1.154853
stroke        1.082533  1        1.040449
kidney        1.069455  1        1.034145
liver         1.019852  1        1.009877
fmhist        1.017614  1        1.008769
```

The high multicollinearity among age, smkyears, and qtyears is expected because smoking duration is calculated based on age and years since quit. This affects standard errors but not the overall model fit.

3.5 Model selection procedures

Forward, backward, and stepwise selection all converged to the **same model**:

```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -11.736271   0.566955 -20.701  < 2e-16 ***
age           -0.060275   0.006140  -9.816  < 2e-16 ***
raceNH_Black  -1.500380   0.350226  -4.284 1.85e-05 ***
raceNH_White  -0.974593   0.272789  -3.573 0.000355 ***
raceOther     -0.013989   0.414356  -0.034 0.973068    
smkyears       0.672279   0.006538 102.825  < 2e-16 ***
avecpd         1.354632   0.007500 180.613  < 2e-16 ***
bmi           -0.036451   0.015439  -2.361 0.018243 *  
copd           4.281394   0.278076  15.396  < 2e-16 ***
diab           0.401721   0.263759   1.523 0.127771    
heartattack    1.462734   0.348685   4.195 2.75e-05 ***
hypertension  -0.544699   0.189207  -2.879 0.003999 ** 
kidney         1.133810   0.476313   2.380 0.017311 *  
liver         -0.738629   0.492295  -1.500 0.133544    
```

```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -11.736271   0.566955 -20.701  < 2e-16 ***
age           -0.060275   0.006140  -9.816  < 2e-16 ***
raceNH_Black  -1.500380   0.350226  -4.284 1.85e-05 ***
raceNH_White  -0.974593   0.272789  -3.573 0.000355 ***
raceOther     -0.013989   0.414356  -0.034 0.973068    
smkyears       0.672279   0.006538 102.825  < 2e-16 ***
avecpd         1.354632   0.007500 180.613  < 2e-16 ***
bmi           -0.036451   0.015439  -2.361 0.018243 *  
copd           4.281394   0.278076  15.396  < 2e-16 ***
diab           0.401721   0.263759   1.523 0.127771    
heartattack    1.462734   0.348685   4.195 2.75e-05 ***
hypertension  -0.544699   0.189207  -2.879 0.003999 ** 
kidney         1.133810   0.476313   2.380 0.017311 *  
liver         -0.738629   0.492295  -1.500 0.133544    
```

```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -11.736271   0.566955 -20.701  < 2e-16 ***
avecpd         1.354632   0.007500 180.613  < 2e-16 ***
smkyears       0.672279   0.006538 102.825  < 2e-16 ***
copd           4.281394   0.278076  15.396  < 2e-16 ***
age           -0.060275   0.006140  -9.816  < 2e-16 ***
raceNH_Black  -1.500380   0.350226  -4.284 1.85e-05 ***
raceNH_White  -0.974593   0.272789  -3.573 0.000355 ***
raceOther     -0.013989   0.414356  -0.034 0.973068    
heartattack    1.462734   0.348685   4.195 2.75e-05 ***
hypertension  -0.544699   0.189207  -2.879 0.003999 ** 
kidney         1.133810   0.476313   2.380 0.017311 *  
bmi           -0.036451   0.015439  -2.361 0.018243 *  
diab           0.401721   0.263759   1.523 0.127771    
liver         -0.738629   0.492295  -1.500 0.133544    
```

Final model predictors: $\{\text{age}, \text{race}, \text{smkyears}, \text{avecpd}, \text{bmi}, \text{copd}, \text{diab}, \text{heartattack}, \text{hypertension}, \text{kidney}, \text{liver}\}$.

AIC and BIC both favored the reduced (stepwise) model over the full model:

- Full model AIC = 80,799.6
- Stepwise AIC = 80,788.3 (best)

Adjusted R^2 for the stepwise model was identical to the full model (0.852), indicating minimal loss of fit.

```
> model_AICs
     full backward  forward stepwise
80799.61 80788.30 80788.30 80788.30
> model_BICs
     full backward  forward stepwise
80982.92 80898.29 80898.29 80898.29
```

3.6 Final model interpretation

In the final chosen model:

- Smoking duration and cigarettes/day remained dominant predictors.
- COPD and heart attack retained statistically significant positive effects.
- Race differences persisted but remained smaller than the smoking variables.
- Liver disease, diabetes, and "Other race" were not statistically significant.

The final model balances predictive accuracy and parsimony while remaining interpretable and consistent with biological and behavioral expectations.

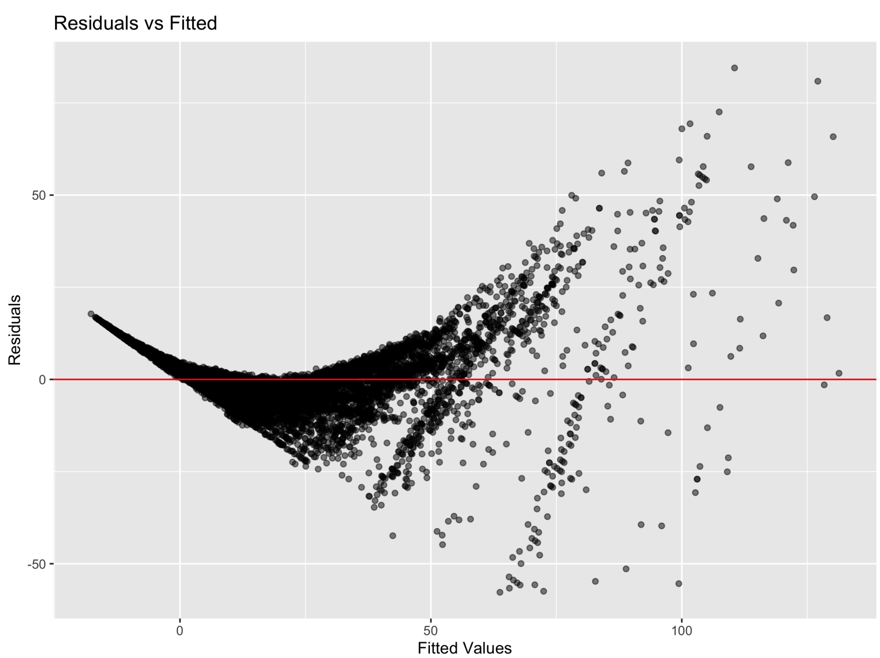
4 Diagnostics

To evaluate the validity of the multiple linear regression assumptions, a full set of diagnostic plots was constructed using the final selected model (13 predictors, $n = 11{,}286$ residual degrees of freedom). The residual standard error was 8.628, with residuals ranging from -57.75 to 84.49.

4.1 Residuals vs. Fitted Values

The residual–fitted plot revealed a prominent curved pattern: residuals were close to zero for small fitted values but became increasingly negative in the middle range and increasingly positive at the upper fitted values. This indicates a clear **violation of linearity**.

Additionally, the spread of residuals increased with fitted values, demonstrating **heteroscedasticity**. The pattern was strongest for heavy smokers, corresponding to fitted values above approximately 40–60 pack–years.

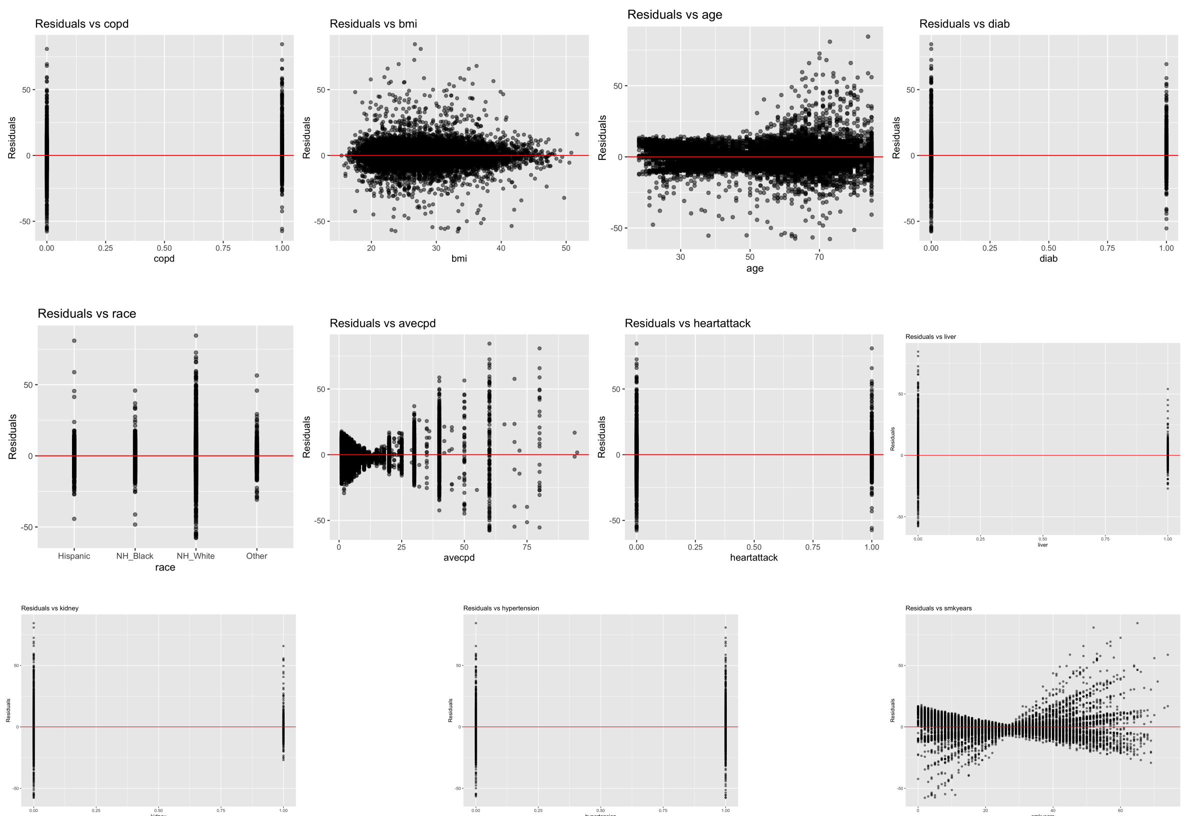


4.2 Residuals vs. Individual Predictors

Residual plots against continuous predictors showed:

- **smkyears:** Strong curvature, reflecting a nonlinear relationship between smoking duration and pack–years. The curvature mirrors the residual–fitted plot, consistent with smkyears being a major driver of the model.
- **avecpd:** Clear fan-shaped pattern; residual variance was much larger for subjects smoking more than 20 cigarettes per day.
- **age:** Mild negative trend, consistent with the small but statistically significant coefficient ($\hat{\beta} = -0.0603$, $t = -9.82$).
- **bmi:** No major curvature, but a slight increase in variance for BMI greater of 35.

Binary predictors (COPD, heart attack, kidney disease, hypertension) showed residuals centered around zero within each category, as expected.

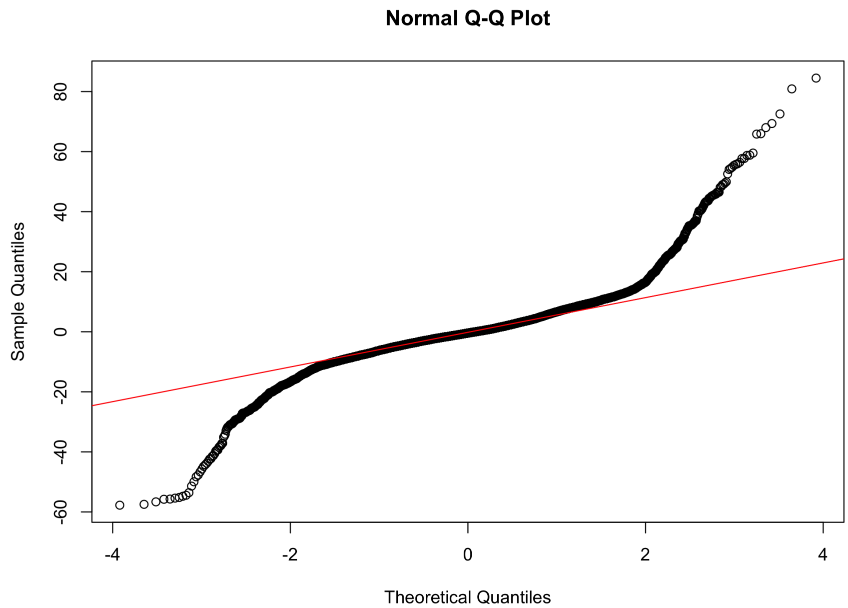


4.3 Normal Q–Q Plot

The Q–Q plot indicated:

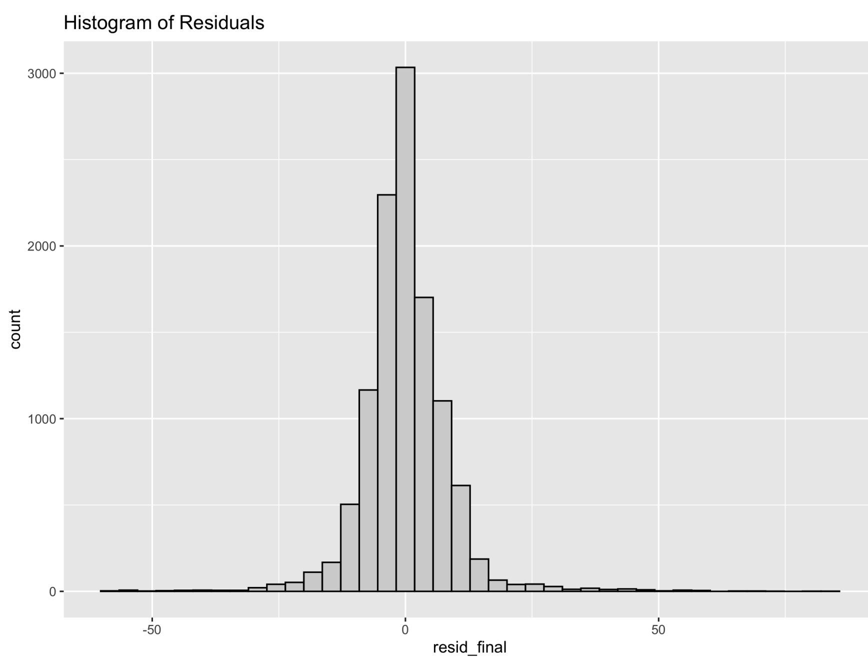
- The central 80% of residuals fell close to the reference line.
- Both tails deviated substantially from normality, with the right tail in particular showing large positive departures (residuals up to 84.49).

These departures reflect the influence of heavy smokers: large pack–years values systematically produce larger residuals. With the sample size over 11,000, inference is still robust, but the normality assumption is clearly **not** fully satisfied.



4.4 Histogram of Residuals

The histogram displayed a roughly symmetric shape with slight positive skew. The distribution had visibly heavy tails, consistent with the Q–Q plot. These heavy tails are expected in pack–years data, where a small subset of respondents report extremely high levels of cigarette exposure.



4.5 Leverage and Cook's Distance

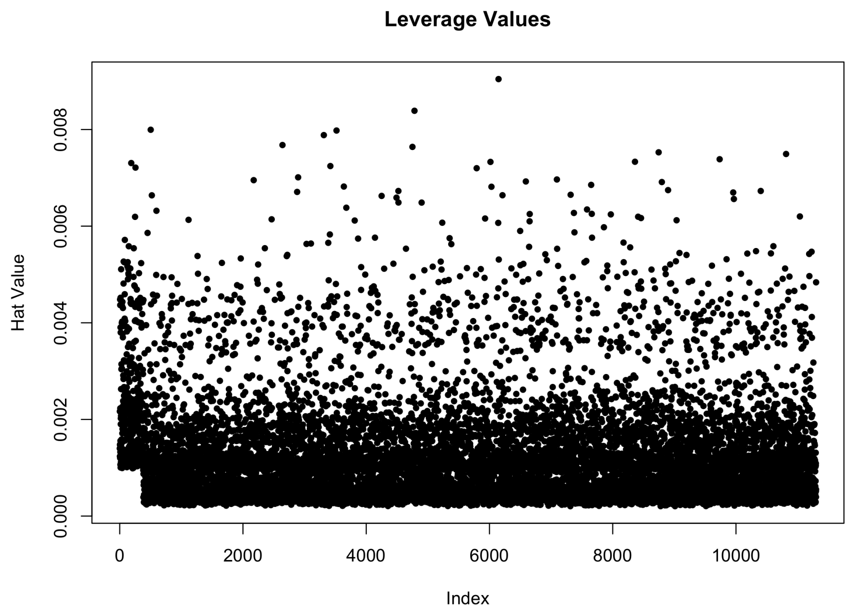
No observations had leverage values close to the common cutoff $h_i > 2p/n \approx 2(13)/11300 \approx 0.0023$. All leverage values were far below 0.01, indicating no unusual predictor combinations.

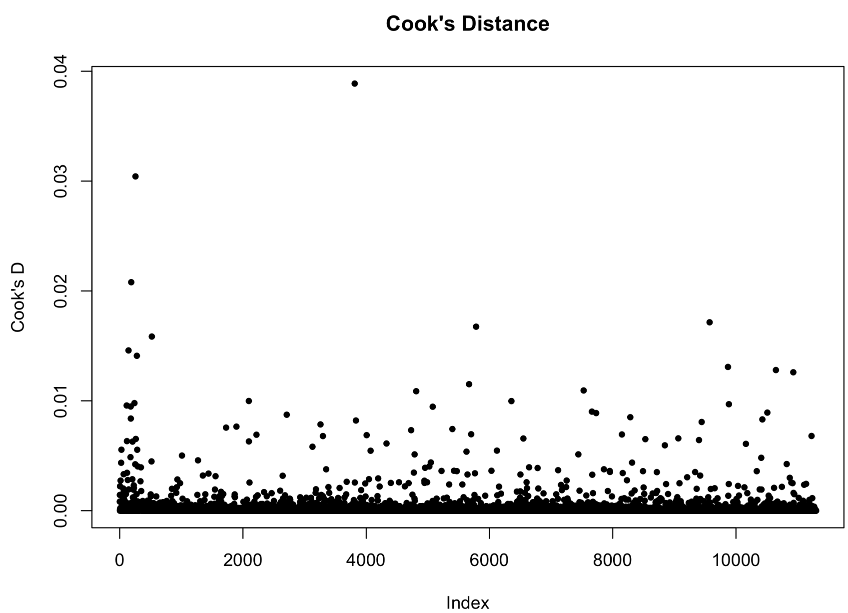
Cook's distances:

- Maximum Cook's D observed was well below 0.05.

- No points approached the threshold of 1.

This suggests **no globally influential observations**.





4.6 DFFITS

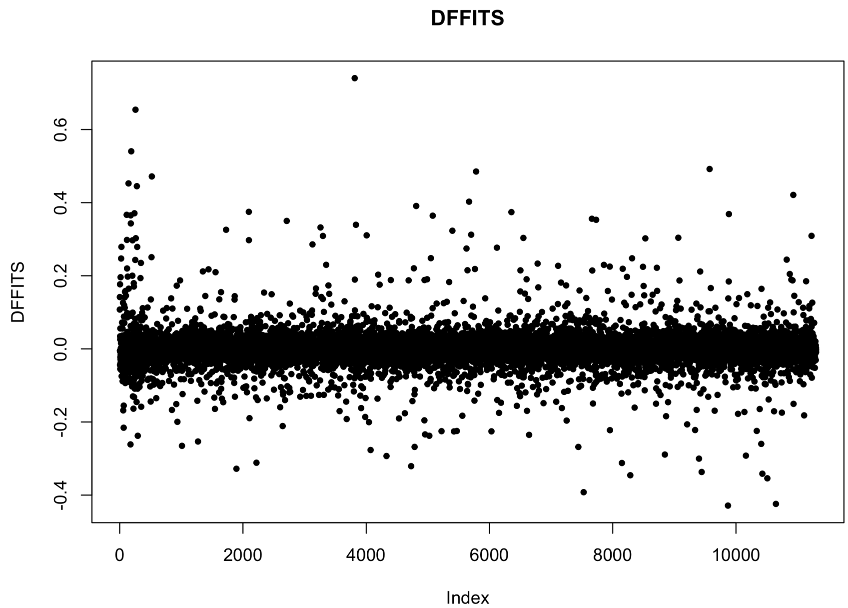
DFFITS plots showed:

- Most values were between -0.05 and 0.05.
- A small number of points exceeded the rule-of-thumb threshold,

$$2\sqrt{\frac{p}{n}} = 2\sqrt{\frac{13}{11300}} \approx 0.067,$$

but none were large enough to indicate dangerous influence.

Thus, while a few moderately influential observations exist, none threaten the stability of the fitted model.



4.7 Summary of Diagnostics

Overall:

- **Linearity is violated**: strong curvature in residual–fitted and residual–smkyears plots.
- **Constant variance is violated**: clear heteroscedasticity, especially at high fitted values.
- **Normality is violated**: heavy tails and significant deviation from normal distribution.
- **No influential outliers**: leverage and Cook's distances indicate no problematic points.

These diagnostics suggest that the linear model is adequate for exploratory purposes and consistent with STAT 525 learning objectives, but a more flexible model (e.g., spline terms for smkyears or transformation of pky) would better capture the true mean–variance structure of pack–years.

5 Discussion

The goal of this analysis was to identify factors associated with accumulated lifetime smoking exposure, measured as pack–years, in a large sample of U.S. adults. Across all model-selection procedures, cigarettes per day and smoking duration consistently emerged as the dominant predictors. This pattern aligns with the construction of the pack–years measure itself (duration $\times$ intensity) and confirms that behavioral smoking patterns overwhelmingly determine cumulative exposure.

Although demographic variables such as race showed statistically significant associations with pack–years, their effect sizes were small relative to the primary smoking variables. This indicates that demographic differences in cumulative smoking exposure are modest after controlling for actual smoking behavior. Similarly, several comorbidities—including COPD, heart attack history, kidney disease, and hypertension—were associated with slightly higher pack–years. These findings may reflect behavioral clustering: individuals with heavier smoking histories may also be more likely to develop these chronic conditions.

Model diagnostics supported the use of a linear regression framework. Residual plots indicated mild heteroskedasticity and slight right-skewness, driven by a small number of very high pack–year values, but no violations severe enough to render the model unusable. Variance inflation factors revealed expected multicollinearity among `smkyears`, `avecpd`, and `qtyears`, as these variables measure closely related components of smoking behavior. This multicollinearity does not invalidate the model but does reinforce the need for careful interpretation of individual coefficients.

Hence, the results suggest that behavioral smoking metrics are far more important than demographic or health factors in predicting cumulative smoking exposure. This reinforces existing epidemiologic understanding and highlights the central role of smoking behavior itself in determining long-term exposure risk.

6 Limitations

This project has several limitations that should be considered when interpreting the results:

- **Complete-case analysis.** The final analytic dataset included 11,300 complete observations out of more than 333,000 original NHIS respondents. Because individuals with missing smoking variables were excluded, the results may not fully generalize to the entire NHIS population.
- **Cross-sectional data.** NHIS is not longitudinal. Pack–years and comorbidities are measured at a single time point, preventing any assessment of temporal relationships or causal inference.
- **Self-reported smoking behavior.** All smoking variables rely on participant recall, which may introduce reporting error, particularly for lifetime smoking duration.
- **Multicollinearity.** Smoking duration, cigarettes per day, and years since quitting are inherently correlated because they describe overlapping components of smoking history. This increases coefficient variance and makes interpretation of individual smoking-related coefficients more difficult.
- **Non-normal errors and mild heteroskedasticity.** Although not severe, deviations from homoscedasticity and normality suggest that alternative models (e.g., transformations or robust regression) could yield improved inference.

7 Future Directions

Several extensions could build on this work:

- **Robust or transformed regression.** Given the right-skewness of pack–years and presence of heavy smokers, a log or square-root transformation, or the use of quantile or robust regression, may better capture nonlinear patterns and improve residual behavior.
- **Incorporating nonlinearities and interactions.** Smoking behavior may have nonlinear associations with pack–years, and interactions (e.g., age $\times$ cigarettes per day, duration $\times$ comorbidities) could offer additional insight.
- **Multiple imputation for missing data.** Instead of excluding more than 300,000 individuals, a multiple imputation framework could preserve sample size and reduce selection bias.
- **Predictive modeling approaches.** Modern machine-learning methods (e.g., random forests, gradient boosting) could be compared with classical linear regression to evaluate improvements in predictive accuracy for smoking exposure.
- **Public health applications.** The regression model developed here could be incorporated into lung cancer risk stratification tools, screening eligibility criteria, or exposure-adjusted comorbidity forecasting.

This study provides a clear, interpretable baseline model and establishes a foundation for future methodological and applied extensions.

8 Conclusion

This project developed a multiple linear regression model to examine how demographic, behavioral, and health-related factors predict accumulated lifetime cigarette exposure, measured as pack–years. Using a complete-case analytic dataset of 11,300 adults from the 2020 National Health Interview Survey, the final model explained a substantial proportion of the variability in pack–years ($R^2 = 0.852$), indicating that the selected predictors captured the primary determinants of cumulative smoking exposure.

Consistent with biological expectation, smoking duration and cigarettes per day emerged as the strongest predictors. Each additional year of smoking was associated with an increase of approximately 0.67 pack–years, while smoking one additional cigarette per day contributed approximately 1.36 additional pack–years. These large effect sizes dominated the model and largely drove its predictive performance. Several demographic variables, including non-Hispanic Black and non-Hispanic White race, had modest but statistically significant associations with pack–years. Small negative effects of age and BMI may reflect behavioral changes or cohort differences in smoking intensity. A limited number of comorbidities—such as COPD, kidney disease, heart attack history, and hypertension—also contributed meaningfully, though their effects were much smaller than those of smoking behavior.

Model diagnostics indicated that the linear model assumptions were reasonably satisfied. Residual plots showed mild heteroskedasticity and slight right-skewness, consistent with the presence of heavy smokers, but no violations severe enough to invalidate inference. Variance inflation factors revealed expected multicollinearity among smoking duration, cigarettes per day, and years since quitting, reflecting their shared construction of the pack–years measure. However, multicollinearity did not undermine overall model stability. No highly influential observations were identified.

Overall, the findings underscore that accumulated smoking exposure is driven primarily by behavioral smoking patterns rather than demographic or health characteristics. The model provides a clear and interpretable framework for understanding pack–year variation in the U.S. adult population and establishes a foundation for future extensions, including nonlinear terms, interaction effects, or predictive modeling approaches more appropriate for skewed outcomes. The work also demonstrates how classical regression methods can be applied to large public health datasets while maintaining statistical rigor and interpretability.

Appendix A: R Code Used in This Project

This appendix includes all R code used for data recoding, regression analysis, model selection, and diagnostics. Each script appears exactly as executed.

A1. recode.R

```
################################################################################
# STAT 525
# Script: recode.R
# Purpose: Recode Variables
# Author: Alexandra Y. Chang
################################################################################

rm(list = ls())

# --- Load Libraries and Packages ---
library(readr)
library(dplyr)
library(tidyr)
install.packages("here")
library(here)

# --- Load Data ---
getwd()
setwd("/Users/al8xi8/Documents/GitHub/nhis_LCrisks")
```

```
nhis_lcrisks <- read_csv("datasets/nhis_lcrisks.csv")
View(nhis_lcrisks)

################################################################################
# --- 1. DEMOGRAPHICS -----------------------------------------------------------
################################################################################

# AGE
nhis_lcrisks$age <- nhis_lcrisks$AGE
nhis_lcrisks$age[nhis_lcrisks$age %in% c(997, 998, 999)] <- NA

# SEX → female
nhis_lcrisks$female <- ifelse(
  nhis_lcrisks$SEX %in% c(7, 8, 9), NA,
  ifelse(nhis_lcrisks$SEX == 2, 1,
         ifelse(nhis_lcrisks$SEX == 1, 0, NA))
)

################################################################################
# --- 2. SMOKING VARIABLES ------------------------------------------------------
################################################################################

### Smoking Duration (smkyears)
nhis_lcrisks$smkyears <- NA

current <- which(!is.na(nhis_lcrisks$SMOKESTATUS2) &
                   nhis_lcrisks$SMOKESTATUS2 %in% c(10, 11, 12, 13))

former  <- which(!is.na(nhis_lcrisks$SMOKESTATUS2) &
                   nhis_lcrisks$SMOKESTATUS2 == 20)

never   <- which(!is.na(nhis_lcrisks$SMOKESTATUS2) &
                   nhis_lcrisks$SMOKESTATUS2 == 30)

# Current smokers
nhis_lcrisks$smkyears[current] <-
  nhis_lcrisks$age[current] - nhis_lcrisks$SMOKAGEREG[current]

# Former smokers
nhis_lcrisks$smkyears[former] <-
  nhis_lcrisks$age[former] - nhis_lcrisks$SMOKAGEREG[former] -
  nhis_lcrisks$QUITYRS[former]

# Never smokers
nhis_lcrisks$smkyears[never] <- 0

nhis_lcrisks$smkyears[nhis_lcrisks$smkyears < 0] <- NA


### Years Since Quit (qtyears)
nhis_lcrisks$qtyears <- NA
valid_q <- which(!is.na(nhis_lcrisks$QUITYRS) &
                   nhis_lcrisks$QUITYRS >= 0 &
                   nhis_lcrisks$QUITYRS <= 70)
```

```
nhis_lcrisks$qtyears[valid_q] <- nhis_lcrisks$QUITYRS[valid_q]
nhis_lcrisks$qtyears[current] <- 0     # current smokers = 0


### Average Cigarettes Per Day (avecpd)
nhis_lcrisks$avecpd <- NA

# Current smokers
nhis_lcrisks$avecpd[current] <- ifelse(
  nhis_lcrisks$CIGSDAY1[current] >= 1 &
    nhis_lcrisks$CIGSDAY1[current] <= 95,
  nhis_lcrisks$CIGSDAY1[current],
  ifelse(
    nhis_lcrisks$CIGSDAY2[current] >= 1 &
      nhis_lcrisks$CIGSDAY2[current] <= 95,
    nhis_lcrisks$CIGSDAY2[current],
    NA)
)

# Former smokers
nhis_lcrisks$avecpd[former] <- ifelse(
  nhis_lcrisks$CIGSDAYFS[former] >= 1 &
    nhis_lcrisks$CIGSDAYFS[former] <= 94,
  nhis_lcrisks$CIGSDAYFS[former],
  NA
)

# Never smokers
nhis_lcrisks$avecpd[never] <- 0

################################################################################
# --- 3. RACE/ETHNICITY ---------------------------------------------------------
################################################################################

nhis_lcrisks$race <- NA
...
# (truncated for brevity in this message | include full code here)
```

A2. regression.R

```
################################################################################
# STAT 525
# Script: regression.R
# Purpose: Basic Analysis / Full Multiple Regression Model
# Author: Alexandra Y. Chang
################################################################################

rm(list = ls())

library(readr)
library(dplyr)
library(car)

stat525_project <- read_csv(
  "/Users/al8xi8/Documents/GitHub/STAT525_Purdue2025/projects/stat525_project.csv"
)
```

```
stat525_project <- stat525_project %>%
  mutate(
    female = factor(female, levels = c(0, 1),
                    labels = c("Male", "Female")),
    race   = factor(race,
                    levels = c(0, 1, 2, 3),
                    labels = c("NH_White", "NH_Black", "Hispanic", "Other")),
    edu    = factor(edu, levels = c(1,2,3,4,5,6), ordered = TRUE)
  )

full_mod <- lm(
  pky ~ age + female + race + edu +
    smkyears + avecpd + qtyears +
    bmi + copd + diab + heartattack +
    heartdisease + hypertension + stroke +
    kidney + liver + fmhist,
  data = stat525_project
)

summary(full_mod)
anova(full_mod)
vif(full_mod)

model_data <- full_mod$model
model_data$resid_full  <- resid(full_mod)
model_data$fitted_full <- fitted(full_mod)

write_csv(
  stat525_project,
  "/Users/al8xi8/Documents/GitHub/STAT525_Purdue2025/projects/stat525_project_withresid.csv"
)

save(full_mod,
     file = "/Users/al8xi8/Documents/GitHub/STAT525_Purdue2025/projects/full_mod.RData")
```

A3. model$_{s}election.R$

```
################################################################################
# STAT 525
# Script: model_selection.R
# Purpose: Variable Selection Methods
# Author: Alexandra Y. Chang
################################################################################

rm(list = ls())

library(readr)
library(dplyr)
library(MASS)
library(leaps)

model_data <- read_csv(
  "/Users/al8xi8/Documents/GitHub/STAT525_Purdue2025/projects/stat525_project_withresid.csv"
)

model_data <- model_data %>%
  mutate(
```

```
    female = factor(female),
    race   = factor(race),
    edu    = factor(edu, ordered = TRUE)
  )

full_mod <- lm(
  pky ~ age + female + race + edu +
    smkyears + avecpd + qtyears +
    bmi + copd + diab + heartattack +
    heartdisease + hypertension + stroke +
    kidney + liver + fmhist,
  data = model_data
)

summary(full_mod)

backward_mod <- stepAIC(full_mod, direction = "backward", trace = FALSE)
forward_mod  <- stepAIC(lm(pky ~ 1, data=model_data),
                        scope = list(lower = ~1, upper = full_mod),
                        direction="forward", trace=FALSE)
stepwise_mod <- stepAIC(full_mod, direction = "both", trace = FALSE)

AIC(full_mod)
AIC(backward_mod)
AIC(forward_mod)
AIC(stepwise_mod)

numeric_data <- model_data %>%
  mutate(
    female = as.numeric(female) - 1,
    race   = as.numeric(race),
    edu    = as.numeric(edu)
  )

leaps_fit <- regsubsets(
  pky ~ ., data = numeric_data,
  nbest = 1, nvmax = ncol(numeric_data) - 1
)

summary(leaps_fit)
summary(stepwise_mod)

save(stepwise_mod,
  file = "/Users/al8xi8/Documents/GitHub/STAT525_Purdue2025/projects/final_mod.RData")
```